



X & O

Function

For & while

If & switch

Variable Type

C++

Eng. Nour Alhamwi





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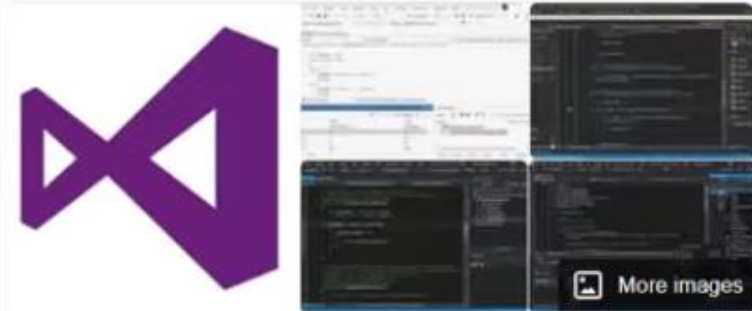
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Computer program

Microsoft Visual Studio is an integrated development environment from Microsoft. It is used to develop computer programs, as well as websites, web apps, web services and mobile apps. [Wikipedia](#)

License: [Freemium](#)

Operating system: [Windows 7 SP1](#) and later; [Windows Server 2012 R2](#) and later; [Mac OS](#)

Developed by: [Microsoft Corporation](#)

Stable release: 2019 version 16.7.0 (16.7.30330.147) (August 5, 2020; 15 days ago)

Preview release: 2019 version 16.8.0 [Preview 1](#) (16.8.30404.54)



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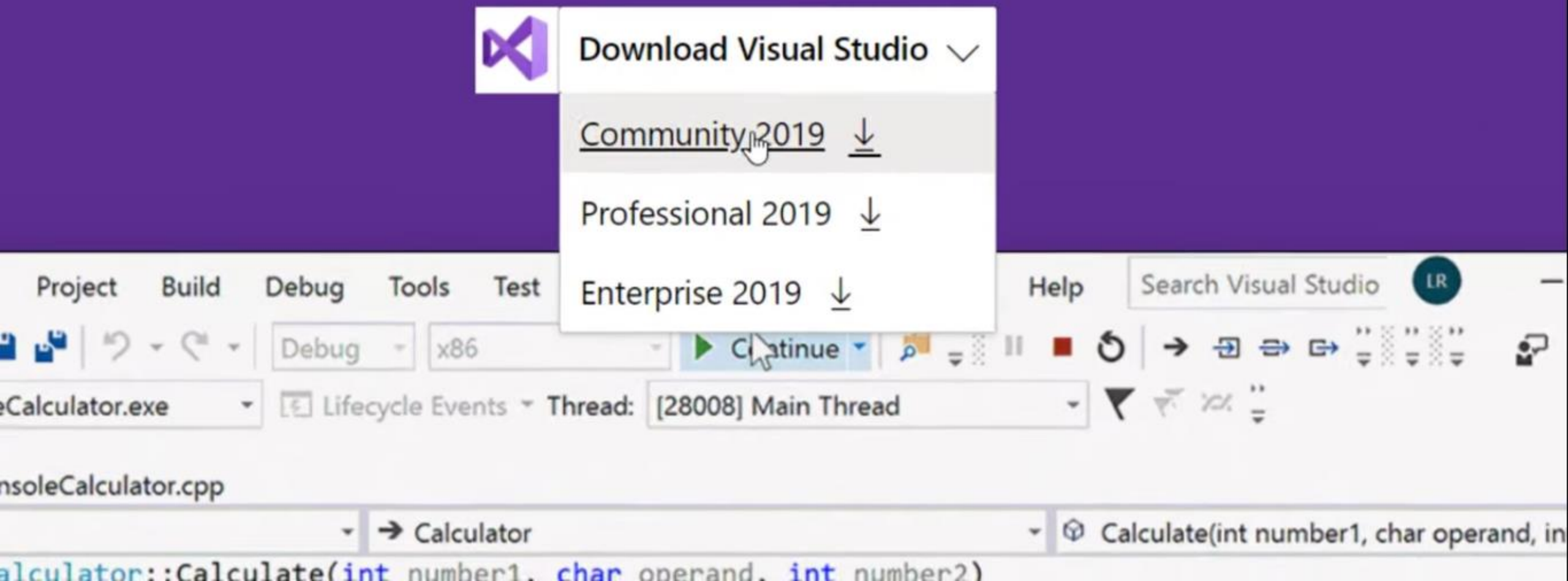
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Workloads Individual components Language packs Installation locations

Web & Cloud (4)

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Build web applications using ASP.NET Core, ASP.NET, HTML/JavaScript, and Containers including Docker support.

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Azure SDKs, tools, and projects for developing cloud apps and creating resources using .NET Core and .NET...

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Build scalable network applications using Node.js, an asynchronous event-driven JavaScript runtime.



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**Desktop development with C++**

Build modern C++ apps for Windows using tools of your choice, including Visual C++, Clang, CMake, or MSBuild.

**Universal Windows Platform development**

Create applications for the Universal Windows Platform with C#, VB, or optionally C++.

**Mobile development with .NET**

Build cross-platform applications for iOS, Android or Windows using Xamarin.



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> Visual Studio core editor

✓ Desktop development with C++

Included

- ✓ C++ core desktop features
- ✓ IntelliCode

Optional

- ☒ MSVC v142 - VS 2019 C++ x64/x86 build tools (...)
- ☒ Windows 10 SDK (10.0.18362.0)
- ☒ Just-In-Time debugger
- ☒ C++ profiling tools
- ☒ C++ CMake tools for Windows
- ☒ C++ ATL for latest v142 build tools (x86 & x64)
- ☒ Test Adapter for Boost.Test
- ☒ Test Adapter for Google Test
- ☒ Live Share
- ☒ C++ AddressSanitizer (Experimental)
- ☐ C++ MFC for latest v142 build tools (x86 & x64)
- ☐ C++/CLI support for v142 build tools (14.27)
- ☐ C++ Modules for v142 build tools (x64/x86 - ex...
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Node.js development

Build scalable network applications using Node.js, an asynchronous event-driven JavaScript runtime.



Desktop development with C++

Build modern C++ apps for Windows using tools of your choice, including MSVC, Clang, CMake, or MSBuild.



Mobile development with .NET

Build cross-platform applications for iOS, Android or Windows using Xamarin.



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Visual Studio Installer

Installed

Available



Visual Studio Community 2019

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100%

Installing... This might take a while.

99%

Optimizing assemblies...

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Restart

Not now

تعليمات أساسية

```
#include<iostream>
```

تضمين المكتبة

```
using namespace std;
```

استخدام فضاء الأسماء std

```
void main()  
{  
  
}
```

main

تعليمات القراءة والكتابة

```
cout << "x=" << x << endl ;
```

الإخراج أو الكتابة

```
cin >> x ;
```

القراءة من واجهة المستخدم

```
#include<iostream>
using namespace std;
void main()
{
    string name;
    cout << "Enter your name" << endl;
    cin >> name;
    cout << "Hello " << name << endl;
}
```

أنواع المتغيرات

اسم النوع	يستعمل لتخزين	مثال
char	أحرف	'a', 'B', '\$', '3', '?'
string	كلمات	"nour"
int	أرقام صحيحة عادي	100 23 2
double	أرقام حقيقية	-0.048512 753393.95

; قيمة = اسم المتغير **نمط المتغير**

int a = 7;

int b = 3, c = 1;

double y;

char Opt;

string name;

Compiler & interpreter

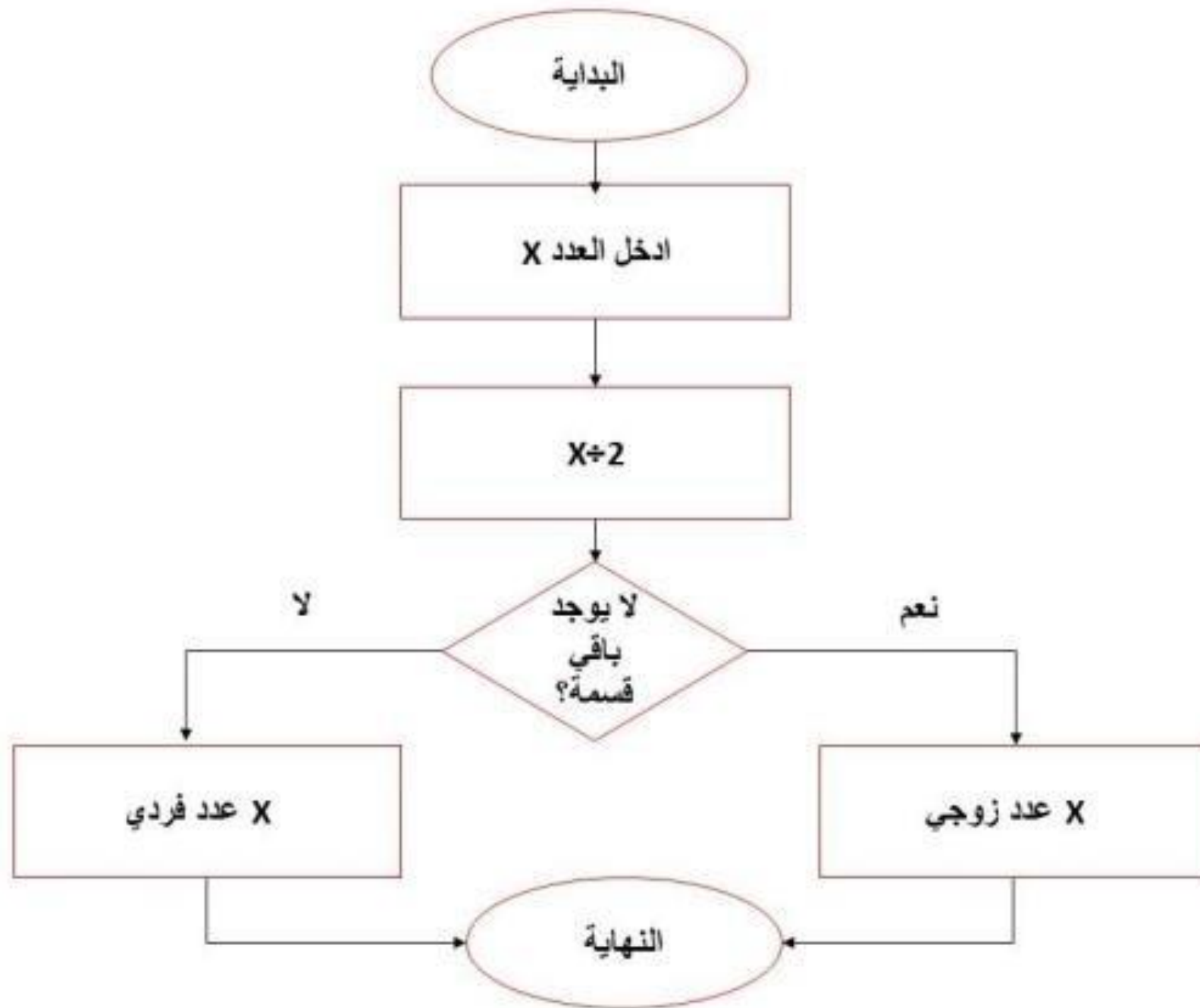
العمليات الحسابية

العملية	الوصف	مثال
+	جمع	$X+Y$
-	طرح	$X-Y$
/	قسمة	$X*Y$
*	ضرب	X/Y
%	باقي القسمة	$X\%Y$
++	زيادة بمقدار 1	$X++$
--	نقصان بمقدار 1	$X--$

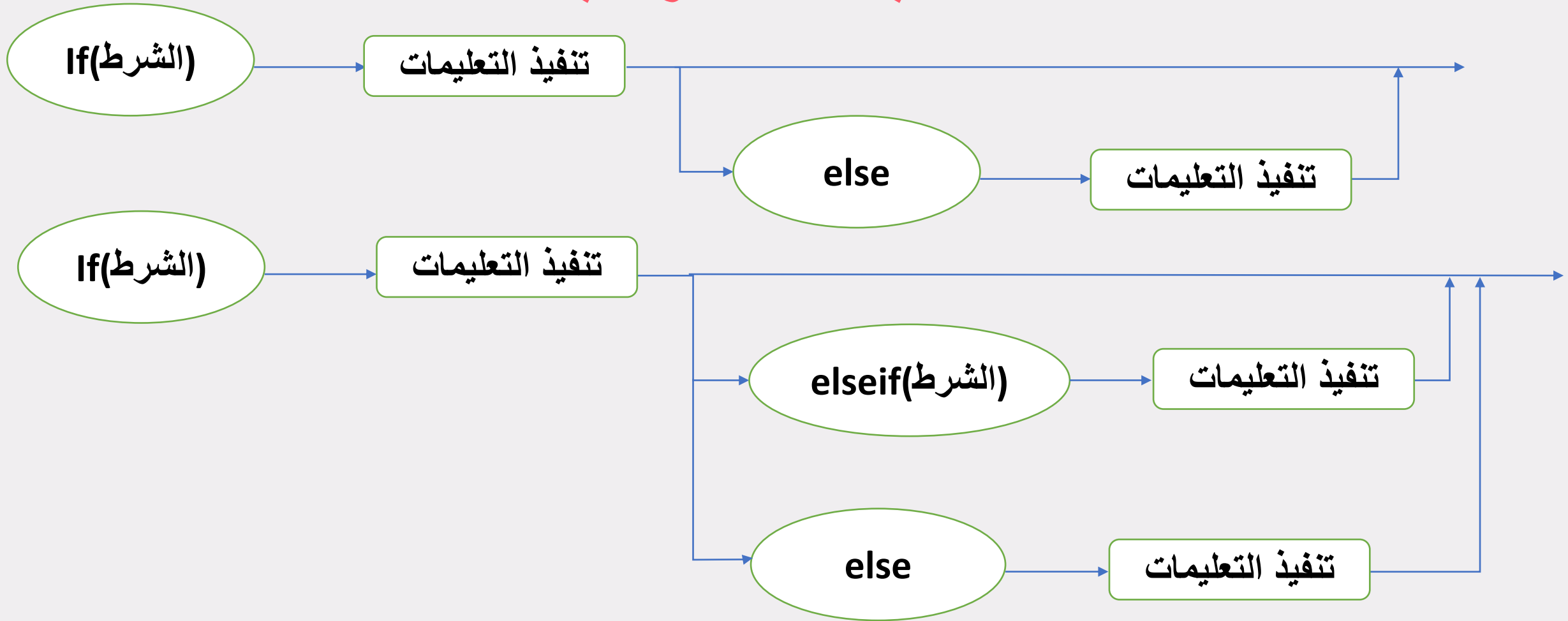
عمليات المقارنة

العملية	الوصف	مثال
>	أكبر تماما	$A>B$
>=	أكبر أو يساوي	$A>=B$
<	أصغر تماما	$A<B$
<=	أصغر أو يساوي	$A<=B$
==	يساوي	$A==B$
!=	لا يساوي	$A!=B$
!	عكس العملية أو نفيها	$!((A+B)<6)$
&&	و	$A>B \ \&\& \ A>C$
	أو	$A>B \ \ A>C$





التعليمة الشرطية if



```
#include<iostream>
using namespace std;
void main()
{
int n;
cout << "Enter number" << endl;
cin >> n;
if (n % 2 == 0)
cout << "even number" << endl;
else
cout << "odd number" << endl;
}
```

مثال if

```

#include<iostream>
using namespace std;

void main()
{
double X, Y;
char Opt;
cout << "inter your X & Opt & Y  " << endl;
cin >> X >> Opt >> Y ;
cout << "-----" << endl;

if (Opt == '+')
cout << X << Opt << Y << " = " << X + Y << endl;

else if (Opt == '-')
cout << X << Opt << Y << " = " << X - Y << endl;

else if (Opt == '*')
cout << X << Opt << Y << " = " << X * Y << endl;

else if (Opt == '/')
cout << X << Opt << Y << " = " << X / Y << endl;

else
cout << "Error Input "<<endl;
}

```

آلة حاسبة باستخدام if

switch

switch(اسم المتحول){

حيث 1,2,3 هي قيم للمتحول

تنفيذ التعليمات: case 1

break;

تنفيذ التعليمات: case 2

break;

تنفيذ التعليمات: case 3

break;

تنفيذ التعليمات: default

break;

}

مثال switch

```
#include<iostream>
using namespace std;
void main()
{
    int n;
    cout << "Enter the number : ";
    cin >> n;
    switch (n)
    {
        case 0:
            cout << "True" << endl; break;
        case 1:
            cout << "False" << endl; break;
        default:
            cout << "Error" << endl;
    }
}
```

التعليمة التكرارية for

for(العداد ; الشرط ; تعريف المتحول)

{تنفيذ التعليمات}

```
#include<iostream>
using namespace std;
void main()
{
for (int i = 1; i <= 25; i++)
{
cout << "Hello ^^" << i << endl;
}
}
```

مثال for

إدخال 10 أرقام والحصول على متوسطهم باستخدام for

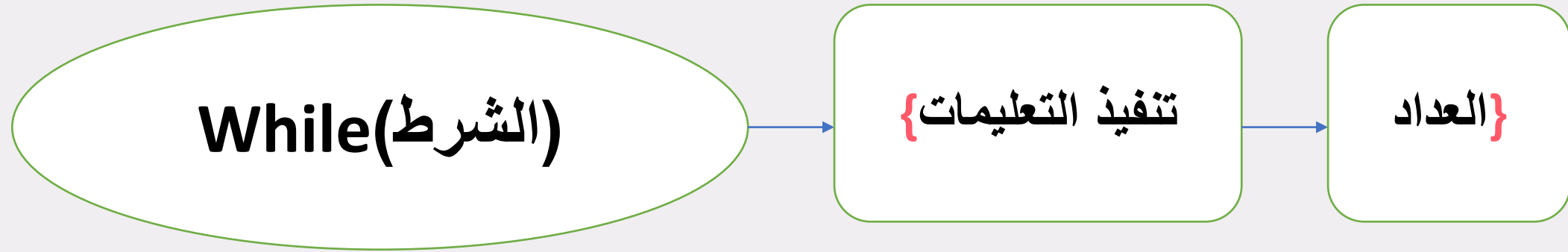
```
#include<iostream>
using namespace std;
void main()
{
double sum = 0;
double avg;
int n;
for (int i = 1; i <= 10; i++) {
cout << "Enter number"<< i << endl;
cin >> n;
}
for (int j = 0; j <= 10; j++) {
sum = sum + n;
}
avg = sum / 10;
cout << "avg =" << avg << endl;
}
```

رسم مستطيل باستخدام for

```
#include<iostream>
using namespace std;
void main()
{

for (int i = 0; i < 7; ++i)
{
for (int j = 0; j < 5; ++j)
cout << "*" << endl;
}
}
```

تعلیمة While



```
#include<iostream>
using namespace std;
void main()
{
    int i = 1;
    while (i <= 10) {
        cout << i << endl;
        i++;
    }
}
```

مثال While

جدول الضرب لعدد واحد باستخدام While

```
#include<iostream>
using namespace std;
void main()
{

int n;
int i = 1;
cout << "Enter number" << endl;
cin >> n;
while (i <= 10) {
cout << n << "*" << i << "=" << n*i<< endl;
i++;
}
}
```

جدول ضرب جميع الاعداد باستخدام While

```
#include<iostream>
using namespace std;
void main()
{

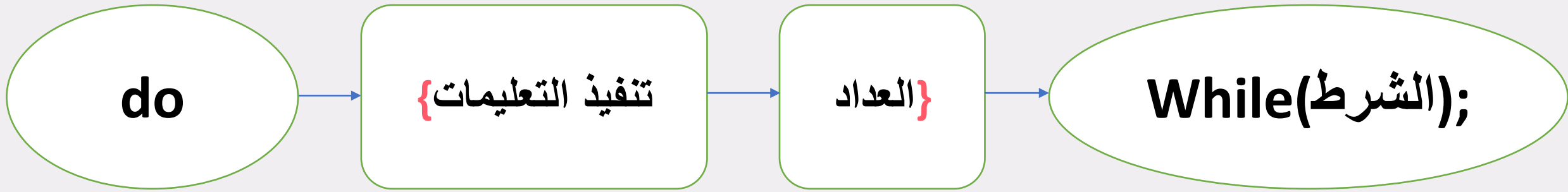
int i = 1;
while (i <= 10) {
int n = 1;
while (n <= 10){
cout <<i << "*" << n << "=" << n * i <<
endl;
n++;
}
cout << "-----" <<
endl;
i++;
}
}
```

تعديل بسيط على خرج جدول الضرب

```
#include<iostream>
using namespace std;
void main()
{

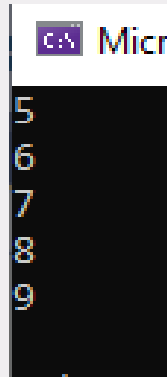
    int i = 1;
    while (i <= 10) {
        int n = 1;
        while (n <= 10){
            cout <<i << "*" << n << "=" << n * i << "||";
            n++;
        }
        cout << endl;
        i++;
    }
}
```

do While تعليمة



break

```
#include<iostream>
using namespace std;
void main()
{
for (int k = 5; k <= 25; ++k)
{
if (k == 10 || k == 20)
break;
cout << k << endl ;
}
}
```

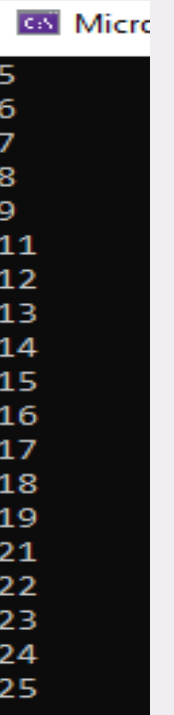


```
C:\Micro
5
6
7
8
9
```

&

continue

```
#include<iostream>
using namespace std;
void main()
{
for (int k = 5; k <= 25; ++k)
{
if (k == 10 || k == 20)
continue;
cout << k << endl ;
}
}
```



```
C:\Micro
5
6
7
8
9
11
12
13
14
15
16
17
18
19
21
22
23
24
25
```


التوابع

```
void main(){
```

استدعاء التابع أي تنفيذه
func (وسطاء);

```
}
```

```
void func ( وسطاء ){
```

تعليمات
طباعة

```
}
```

أو

```
int func ( وسطاء ){
```

تعليمات

يعيد قيمة

```
return قيمة;
```

```
}
```

مثال عن التابع

```
#include<iostream>
using namespace std;
void myMethod()
{
    cout << " Welcome to My Method " << endl;
}
void main() {
    for (int i = 1; i <= 10; i++)
        myMethod();
}
```

X & O

```

#include<iostream>
using namespace std;

char c1 = '1';
char c2 = '2';
char c3 = '3';
char c4 = '4';
char c5 = '5';
char c6 = '6';
char c7 = '7';
char c8 = '8';
char c9 = '9';
void draw() {
    cout << "\t" << c1 << "\t | \t" << c2 << "\t | \t" << c3 << endl;
    cout << "\t----- \n";
    cout << "\t" << c4 << "\t | \t" << c5 << "\t | \t" << c6 << endl;
    cout << "\t----- \n";
    cout << "\t" << c7 << "\t | \t" << c8 << "\t | \t" << c9 << endl;
    cout << endl;
}
int main()
{
    cout << "Welcome to the game X-0" << endl;
    draw();
}

```

دالة رسم شكل اللعبة

```
#include<iostream>
using namespace std;

int input;

void replace(int i)
{
    switch (i)
    {
        case 1:
            c1 = 'x';
            break;
    }
}

int main()
{
    cout << "Welcome to the game X-O" << endl << endl;
    draw();
    cout << "Enter number";
    cin >> input;
    replace(input);
    draw();
}
```

تجريب دالة استبدال الرقم 1
ب X

دالة تبديل الرقم ب c حيث c هي x او o حسب اللاعب

```
void replace(int i, char c)
{
    switch (i)
    {
        case 1: c1 = c; break;
        case 2: c2 = c; break;
        case 3: c3 = c; break;
        case 4: c4 = c; break;
        case 5: c5 = c; break;
        case 6: c6 = c; break;
        case 7: c7 = c; break;
        case 8: c8 = c; break;
        case 9: c9 = c; break;
    }
}
```

```
int main()
{
    int input;
    int c = 0;
    cout << "Welcome to the game X-O" << endl;
    draw();
    while (c<9)
    {
        cout << "Enter number";
        cin >> input;
        if (c % 2 == 0)
            replace(input, 'X');
        else
            replace(input, 'O');
        c++;
        draw();
    }
}
```

غير
ضروري

دالة للتأكد من أن الرقم غير مستخدم سابقا من احد اللاعبين

```
bool check(int input)
{
    switch (input)
    {
        case 1: if (c1 == '1') return true; break;
        case 2: if (c2 == '2') return true; break;
        case 3: if (c3 == '3') return true; break;
        case 4: if (c4 == '4') return true; break;
        case 5: if (c5 == '5') return true; break;
        case 6: if (c6 == '6') return true; break;
        case 7: if (c7 == '7') return true; break;
        case 8: if (c8 == '8') return true; break;
        case 9: if (c9 == '9') return true; break;
    }
    return false;
}
```

```
int main()
{
    int c = 0;
    cout << "Welcome to the game X-O" << endl;
    draw();
    while (c<9)
    {
        cout << "Enter number";
        cin >> input;

        while (!check(input)) {
            cout << "The number is in use, Enter a new number";
            cin >> input;
        }
        if (c % 2 == 0)
        {
            replace(input, 'X');
        }
        else
        {
            replace(input, 'O');
        }
        c++;
        draw();
    }
}
```

```

bool checkWinner()
{
    // التأكد أن الصفوف أحدها تساوي ال X
    bool rox1 = (c1 == 'X' && c2 == 'X' && c3 == 'X');
    bool rox2 = (c4 == 'X' && c5 == 'X' && c6 == 'X');
    bool rox3 = (c7 == 'X' && c8 == 'X' && c9 == 'X');

    // التأكد أن الإعمدة أحدها تساوي ال X
    bool clx1 = (c1 == 'X' && c4 == 'X' && c7 == 'X');
    bool clx2 = (c2 == 'X' && c5 == 'X' && c8 == 'X');
    bool clx3 = (c3 == 'X' && c6 == 'X' && c9 == 'X');

    // التأكد أن القطران أحدها تساوي ال X
    bool dix1 = (c1 == 'O' && c5 == 'O' && c9 == 'O');
    bool dix2 = (c3 == 'O' && c5 == 'O' && c7 == 'O');

    // التأكد أن الصفوف أحدها تساوي ال o
    bool roo1 = (c1 == 'O' && c2 == 'O' && c3 == 'O');
    bool roo2 = (c4 == 'O' && c5 == 'O' && c6 == 'O');
    bool roo3 = (c7 == 'O' && c8 == 'O' && c9 == 'O');

    // التأكد أن الإعمدة أحدها تساوي ال o
    bool clo1 = (c1 == 'O' && c4 == 'O' && c7 == 'O');
    bool clo2 = (c2 == 'O' && c5 == 'O' && c8 == 'O');
    bool clo3 = (c3 == 'O' && c6 == 'O' && c9 == 'O');

    // التأكد أن القطران أحدها تساوي ال o
    bool dio1 = (c1 == 'O' && c5 == 'O' && c9 == 'O');
    bool dio2 = (c3 == 'O' && c5 == 'O' && c7 == 'O');

    if (rox1 || rox2 || rox3 || clx1 || clx2 || clx3 || dix1 || dix2){
        cout << " X is the winner " << endl;
        return true;}
    else if (roo1 || roo2 || roo3 || clo1 || clo2 || clo3 || dio1 || dio2){
        cout << " O is the winner " << endl;
        return true;}
    return false;
}

```

التحقق من الفائز

```

if(checkWinner())
    break;

```

فقط نستدعي التابع في الmain

اختصار خطوات اللعبة

1. يجب علينا رسم الشكل وتعريف محارف في كل فراغ لتبديله فيما بعد ب x, o
2. تبديل الرقم الذي يتم إدخاله ب x, o
3. التأكد من أن الفراغ غير محجوز من قبل
4. إيجاد من الفائز